

json-formula Specification

PDF Association Forms Technical Working Group

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Scope

This document is the specification for json-formula, an expression grammar that operates on JSON (JavaScript Object Notation) documents. The referenced JSON documents and JSON literals must conform to [RFC 8259](#).

The grammar borrows from

- [OpenFormula](#) for spreadsheet operators and function
- [JMESPath](#) for JSON query semantics

The intended audience are both end-users of json-formula as well as implementers; the contents are then both a user guide and a specification.

1. Notation

In the specification, examples are shown through the use of a **search** function called **eval**. The syntax for this function is:

```
eval(<json-formula expr>, <JSON document>) -> <return value>
```

For simplicity, the json-formula expression and the JSON document are not quoted. For example:

```
eval(foo, {"foo": "bar"}) -> "bar"
```

The result of applying a json-formula expression against a JSON document will result in valid JSON, provided there are no errors during the evaluation process.

2. Data Types

json-formula supports all the JSON types:

- number: All numbers are internally represented as double-precision floating-point
- string
- boolean: **true** or **false**
- array: an ordered, sequence of values
- object: an unordered collection of key value pairs

- `null`

There is an additional type that is not a JSON type that's used in json-formula functions:

- expression: A string prefixed with an ampersand (&) character

This specification uses the term "scalar" to refer to any value that is not an array, object or expression. Scalars include numbers, strings, booleans, and null values.

2.1. Type Coercion

If the supplied data is not correct for the execution context, json-formula will attempt to coerce the data to the correct type. Coercion will occur in these contexts:

- Operands of the concatenation operator (&) shall be coerced to a string, except when an operand is an array. Arrays shall be coerced to an array of strings.
- Operands of numeric operators (+, -, *, /) shall be coerced to numbers except when the operand is an array. Arrays shall be coerced to an array of numbers.
- Operands of the union operator (~) shall be coerced to an array
- The left-hand operand of ordering comparison operators (>, >=, <, <=) must be a string or number. Any other type shall be coerced to a number.
- If the operands of an ordering comparison are different, they shall both be coerced to a number
- Parameters to functions shall be coerced when there is a single viable coercion available. For example, if a null value is provided to a function that accepts a number or string, then coercion shall not happen, since a null value can be coerced to both types. Conversely if a string is provided to a function that accepts a number or array of numbers, then the string shall be coerced to a number, since there is no supported coercion to convert it to an array of numbers.

The equality and inequality operators (=, ==, !=, <>) do **not** perform type coercion. If operands are different types, the values are considered not equal.

If an ordering comparison requires coercion, and the coercion is not possible (including the case where a string cannot be coerced to a number), the comparison will return false. e.g., `{a: 12} < 12` and `"12a" < 13` will each return `false`.

In all cases except ordering comparison, if the coercion is not possible, a `TypeError` error shall be raised.

Examples

```
eval("abc" & 123, {}) -> "abc123"
eval("123" * 2, {}) -> 246
eval([1,2,3] ~ 4, {}) -> [1,2,3,4]
eval(123 < "124", {}) -> true
eval("23" > 111, {}) -> false
eval(avgA(["2", "3", "4"]), {}) -> 3
eval(1 == "1", {}) -> false
```

2.2. Type Coercion Rules

Provided Type	Expected Type	Result
number	string	number converted to a string, following the JavaScript <code>toString()</code> rules.
boolean	string	"true" or "false"
array	string	Not supported
object	string	Not supported
null	string	"" (empty string)
string	number	Parse the string to a number. An empty string converts to zero. If the string is not a well-formed number, the coercion will fail.
boolean	number	true → 1 false → 0
array	number	Not supported
object	number	Not supported
null	number	0
number	array	create a single-element array with the number
string	array	create a single-element array with the string
boolean	array	create a single-element array with the boolean
object	array	Not supported

Provided Type	Expected Type	Result
null	array	Not supported
number	object	Not supported
string	object	Not supported
boolean	object	Not supported
array	object	Not supported. Use: <code>fromEntries(entries(array))</code>
null	object	Not supported
number	boolean	zero is false, all other numbers are true
string	boolean	Empty string is false, populated strings are true
array	boolean	Empty array is false, populated arrays are true
object	boolean	Empty object is false, populated objects are true
null	boolean	false

An array may be coerced to another type of array as long as there is a supported coercion for the array content. For example, just as a string can be coerced to a number, an array of strings may be coerced to an array of numbers.

Note that while strings, numbers and booleans may be coerced to arrays, they may not be coerced to a different type within that array. For example, a number cannot be coerced to an array of strings—even though a number can be coerced to a string, and a string can be coerced to an array of strings.

Examples

```
eval("\$123.00\" + 1", {}) -> TypeError
eval("truth is " & `true`, {}) -> "truth is true"
eval(2 + `true`, {}) -> 3
eval(minA(["20", "30"]), {}) -> 20
```

2.3. Date and Time Values

In order to support date and time functions, json-formula represents date and time values as numbers where:

- The integral portion of the number represents the number of days since the epoch: January 1, 1970, [UTC](#)
- The fractional portion of the number represents the fractional portion of the day
- The date/time value is offset from the current time zone to UTC
- The current time zone is determined by the host operating system

The preferred ways to create a date/time value are by using one of these functions:

- `datetime()`
- `now()`
- `toDate()`
- `today()`
- `time()`

Functions that operate on a date/time value will convert the date/time value back to the local time zone.

Examples

```
eval(datetime(1970,1,2,0,0,0) - datetime(1970,1,1,0,0,0), {}) -> 1
eval(datetime(2010,1,21,12,0,0) |
  {month: month(@), day: day(@), hour: hour(@)}, {}) ->
  {"month": 1, "day": 21, "hour": 12}
```

2.4. Integers

Some functions and operators accept numeric parameters that are expected to be integers. In these contexts, if a non-numeric or non-integer value is provided, it will be coerced to a number and then truncated. The specific truncation behaviour is to remove any fractional value without rounding.

2.5. Floating Point Precision

Numbers are represented in [double-precision floating-point format](#). As with any system that uses this

level of precision, results of expressions may be off by a tiny fraction. e.g. $10 * 1.44 \rightarrow 14.399999999999999$

Authors should mitigate this behavior:

- When comparing fractional results, do not compare for exact equality. Instead, compare within a range. e.g.: instead of: `a == b`, use: `abs(a-b) < 0.000001`
- leverage the built-in functions that manipulate fractional values:
 - `ceil()`
 - `floor()`
 - `round()`
 - `trunc()`

3. Errors

Errors may be raised during the json-formula evaluation process. The following errors are defined:

- **EvaluationError** is raised when an unexpected runtime condition occurs. For example, divide by zero.
- **FunctionError** is raised when an unknown function is encountered or when a function receives invalid arguments.
- **SyntaxError** is raised when the supplied expression does not conform to the json-formula grammar.
- **TypeError** is raised when coercion is required for the current evaluation context, but the coercion is not possible.

4. Grammar

The grammar is specified using [Antlr](#). For a machine-readable version of the grammar, see the `grammar.g4` file in the source repository.

```
grammar JsonFormula;

formula : expression EOF ;

expression
  : '(' expression ')'
  | expression bracketExpression
  | bracketExpression
```



```

| objectExpression
| expression '.' chainedExpression
| '!' expression
| '-' expression
| expression ('*' | '/') expression
| expression ('+' | '-' | '~') expression
| expression '&' expression
| expression COMPARATOR expression
| expression '&&' expression
| expression '||' expression
| expression '|' expression
| identifier
| wildcard
| arrayExpression
| JSON_LITERAL
| functionExpression
| STRING
| (REAL_OR_EXPONENT_NUMBER | INT)
| currentNode
;

chainedExpression
: identifier
| arrayExpression
| objectExpression
| functionExpression
| wildcard
;

wildcard : '*' ;

arrayExpression : '[' expression (',' expression)* ']' ;

objectExpression
: '{' keyvalExpr (',' keyvalExpr)* '}'
;

keyvalExpr : identifier ':' expression ;

bracketExpression
: '[' '*' ']'
| '[' slice ']'
| '[' ']'
| '['? expression ']'
| '[' signedInt ']'
;

slice : start=signedInt? ':' stop=signedInt? (':' step=signedInt?)? ;

COMPARATOR
: '<'
| '<='
| '=='
| '='
| '>='

```

```

| '>'
| '!= '
| '<>'
;

functionExpression
: NAME '(' functionArg (',' functionArg)* ')'
| NAME '(' ')'
;

functionArg
: expression
| expressionType
;

currentNode : '@' ;

expressionType : '&' expression ;

identifier
: NAME
| QUOTED_NAME
;

signedInt
: '-?' INT+
;

NAME : [a-zA-Z_$] [a-zA-Z0-9_$]* ;

QUOTED_NAME : '\'' (ESC | ~ ['\\'])* '\'';

JSON_LITERAL
: '"' (ESC | ~ [\\`]) * '"'
;

STRING : '"' (ESC | ~ [ "\\"] ) * '"' ;

fragment ESC : '\\' (UNICODE | [bfnr t\\`"/]);

fragment UNICODE
: 'u' HEX HEX HEX HEX
;

fragment HEX
: [0-9a-fA-F]
;

REAL_OR_EXPONENT_NUMBER
: INT? '.' [0-9] + EXP?
| INT EXP
;

INT
: [0-9]+

```

```

;

fragment EXP
: [Ee] [+|-]? INT
;

WS
: [ \t\n\r] + -> skip
;

```

4.1. Operator Precedence

The antlr grammar defines operator precedence by the order of expressions in the grammar. These are the operators listed from strongest to weakest binding:

- Parenthesis `()`
- Bracket Expression `[...]`
- Braces `{}`
- Dot `.` (chained expressions)
- Flatten: `[]`
- Unary not `!`, unary minus: `-`
- multiply: `*`, divide: `/`
- add: `+`, subtract: `-`, union: `~`
- concatenate: `&`
- Comparison operators: `=`, `==`, `>`, `<`, `>=`, `<=`, `!=`, `<>`
- and: `&&`
- or: `||`
- pipe: `|`

5. Literals

5.1. JSON Literals

```
JSON_LITERAL: '"' (ESC | ~ [\\`]+)* '"';
```

A JSON literal expression allows arbitrary JSON objects to be specified anywhere an expression is

permitted. Implementations should use a JSON parser to parse these literals. Note that the backtick character (```) character must be escaped in a JSON literal which means implementations need to handle this case before passing the resulting string to a JSON parser.

Examples

```
eval(`"foo"`, {}) -> "foo"
eval(`"foo\`bar"`, {}) -> "foo`bar"
eval(`[1, 2]`, {}) -> [1, 2]
eval(`true`, {}) -> true
eval(`{"a": "b"}`.a, {}) -> "b"
eval({first: a, type: `mytype`, {"a": "b", "c": "d"}})
  -> {"first": "b", "type": "mytype"}
```

5.2. String Literals

```
STRING : '"' (ESC | ~["\\])* '"' ;

fragment ESC : '\\' (UNICODE | [bfnrt\\`'"/]);

fragment UNICODE
  : 'u' HEX HEX HEX HEX
  ;

fragment HEX
  : [0-9a-fA-F]
  ;
```

A **STRING** literal is a value enclosed in double quotes and supports the same character escape sequences as strings in **JSON**, as encoded by the **ESC** fragment. e.g., a character 'A' could be specified as the unicode sequence: `\u0041`.

A string literal can also be expressed as a JSON literal. For example, the following expressions both evaluate to the same value: "foo"

```
eval(`"foo"`, {}) -> "foo"
eval("foo", {}) -> "foo"
```

5.3. Number literals

```
numberLiteral = REAL_OR_EXPONENT_NUMBER | INT
```

```
REAL_OR_EXPONENT_NUMBER  
  : INT? '.' [0-9] + EXP?  
  | INT EXP  
  ;
```

```
INT  
  : [0-9]+  
  ;
```

```
fragment EXP  
  : [Ee] [+\-]? INT  
  ;
```

Number literals follow the same syntax rules as numeric values in JSON with three exceptions:

1. Number literals may omit a leading zero before the decimal point. For example, **.123** is not valid JSON, but is allowed as a number literal.
2. Number literals may include leading zeros ahead of the integral part of the number. For example, **0123** is not valid JSON, but is allowed as a number literal.
3. The grammar construction for a number literal does not include a minus sign. Literal expressions are made negative by prefixing them with a unary minus.

Note that number literals (and JSON numbers) allow scientific notation.

Examples

```
eval(44, {}) -> 44  
eval([12, 13], {}) -> [12, 13]  
eval({a: 12, b: 13}, {}) -> {"a": 12, "b": 13}  
eval(foo | [1], {"foo": [3,4,5]}) -> 4  
eval(foo | @[-1], {"foo": [3,4,5]}) -> 5  
eval(foo | [1, 2], {"foo": [3,4,5]}) -> [1, 2]  
eval(6 / 3, {}) -> 2  
eval(1e2, {}) -> 100
```

6. Identifiers

```
identifier
: NAME
| QUOTED_NAME
;

NAME : [a-zA-Z_$] [a-zA-Z0-9_$]* ;

QUOTED_NAME : '\'' (ESC | ~ ['\\])* '\'';
```

An **identifier** is the most basic expression and can be used to extract a single element from a JSON document. The return value for an **identifier** is the value associated with the identifier. If the **identifier** does not exist in the JSON document, then a **null** value is returned.

Using the **NAME** token grammar rule, identifiers can be one or more characters, and must start with a character in the range: **A-Za-z_\$**.

When an identifier has a character sequence that does not match a **NAME** token, an identifier shall follow the **QUOTED_NAME** token rule where an identifier is specified with a single quote ('), followed by any number of characters, followed by another single quote. Any valid string can be placed between single quotes, including **JSON** escape sequences.

Examples

```
eval(foo, {"foo": "value"}) -> "value"
eval(bar, {"foo": "value"}) -> null
eval(foo, {"foo": [0, 1, 2]}) -> [0, 1, 2]
eval('with space', {"with space": "value"}) -> "value"
eval('special chars: !@#"', {"special chars: !@#": "value"}) -> "value"
eval('quote\'char', {"quote'char": "value"}) -> "value"
eval('\u2713', {"\u2713": "value"}) -> "value"
```

7. Operators

7.1. Comparison Operators

The following comparison operators are supported:

- **=**, **==**: test for equality
- **!=**, **<>**: test for inequality
- **<**: less than

- `<=`: less than or equal to
- `>`: greater than
- `>=`: greater than or equal to

7.1.1. Equality Operators

Two representations of the equality and inequality operators are supported: `=` and `==` are equivalent in functionality. Both variations are supported in order to provide familiarity to users with experience with similar grammars. Similarly, `!=` and `<>` function identically. Note that there is no ambiguity with the `=` operator, since json-formula does not have an assignment operator.

- A `string` is equal to another `string` if they have the exact same sequence of code points
- `number` values are compared for an exact match. When comparing fractional values, authors should take into account [floating point precision](#) considerations.
- The literal values `true/false/null` are equal only to their own literal values
- Two JSON objects are equal if they have the same set of keys and values. Two JSON objects `x` and `y`, are considered equal if they have the same number of key/value pairs and if, for each key value pair `(i, j)` in `x`, there exists an equivalent pair `(i, j)` in `y`
- Two JSON arrays are equal if they have equal elements in the same order. Two arrays `x` and `y` are considered equal if they have the same length and, for each `i` from `0` until `length(x)`, `x[i] == y[i]`
- The comparison of array and objects is a deep comparison. That is, where nested arrays or objects are found, the nested elements will be included in the comparison.

7.1.2. Ordering Operators

Ordering comparisons follow these rules:

- If both operands are numbers, a numeric comparison is performed
- If both operands are strings, they are compared as strings, based on the values of the Unicode code points they contain
- If operands are mixed types, [type coercion](#) to number is applied before performing the comparison

7.2. Numeric Operators

The following operators work with numeric operands:

- addition: `+`

- subtraction: $-$
- multiplication $*$
- division: $/$

```
eval(left + right, {"left": 8, "right": 12 }) -> 20
eval(right - left - 10, {"left": 8, "right": 12 }) -> -6
eval(4 + 2 * 4, {}) -> 12
eval(10 / 2 * 3, {}) -> 15
```

7.3. Concatenation Operator

The concatenation operator ($\&$) takes two string operands and combines them to form a single string.

```
eval(left & value & right,
    {"left": "[", "right": "]", "value": "abc" }) -> "[abc]"
eval(map(values, &"$" & @), {"values": [123.45, 44.32, 99.00] }) ->
    ["$123.45", "$44.32", "$99"]
```

7.4. Array Operands

The numeric and concatenation operators ($+$, $-$, $*$, $/$, $\&$) have special behavior when applied to arrays.

- When both operands are arrays, a new array is returned where the elements are populated by applying the operator on each element of the left operand array with the corresponding element from the right operand array
- If both operands are arrays and they do not have the same size, the shorter array is padded with null values
- If one operand is an array and one is a scalar value, the scalar operand will be converted to an array by repeating the value to the same size array as the other operand

```
eval([1,2,3] + [2,3,4], {}) -> [3,5,7]
eval([1,2,3,4] * [1,2,3], {}) -> [1,4,9,0]
eval([1,2,3,4] & "%", {}) -> ["1%", "2%", "3%", "4%"]
```

7.4.1. Union Operator

The union operator ($\sim$) returns an array formed by concatenating the contents of two arrays.


```
eval(a ~ b, {"a": [0,1,2], "b": [3,4,5]}) -> [0,1,2,3,4,5]
eval(a ~ b, {"a": [[0,1,2]], "b": [[3,4,5]]}) -> [[0,1,2],[3,4,5]]
eval(a[] ~ b[], {"a": [[0,1,2]], "b": [[3,4,5]]}) -> [0,1,2,3,4,5]
eval(a ~ 10, {"a": [0,1,2]}) -> [0,1,2,10]
eval(a ~ `null`, {"a": [0,1,2]}) -> [0,1,2,null]
```

7.5. Boolean Operators

7.5.1. Or Operator

The OR operator (`||`) will evaluate to either the left operand or the right operand. If the left operand can be coerced to a true value, it is used as the return value. If the left operand cannot be coerced to a true value, then the right operand is used as the return value. If the left operand is a truth-like value, then the right operand is not evaluated. For example, the expression: `if()` will result in a `FunctionError` (missing arguments), but the expression `true() || if()` will not result in a `FunctionError` because the right operand is not evaluated.

The following conditions cannot be coerced to true:

- Empty array: `[]`
- Empty object: `{}`
- Empty string: `""`
- False boolean: `false`
- Null value: `null`
- zero value: `0`

Examples

```
eval(foo || bar, {"foo": "foo-value"}) -> "foo-value"
eval(foo || bar, {"bar": "bar-value"}) -> "bar-value"
eval(foo || bar, {"foo": "foo-value", "bar": "bar-value"}) -> "foo-value"
eval(foo || bar, {"baz": "baz-value"}) -> null
eval(foo || bar || baz, {"baz": "baz-value"}) -> "baz-value"
eval(override || myarray[-1], {"myarray": ["one", "two"]}) -> "two"
eval(override || myarray[-1], {"myarray": ["one", "two"], "override": "yes"})
  -> "yes"
```

7.5.2. And Operator

The AND operator (`&&`) will evaluate to either the left operand or the right operand. If the left operand

is a truth-like value, then the right operand is returned. Otherwise the left operand is returned. This reduces to the expected truth table:

LHS	RHS	Result
True	True	True
True	False	False
False	True	False
False	False	False

This is the standard truth table for a [logical conjunction](#).

If the left operand is not a truth-like value, then the right operand is not evaluated. For example, the expression: `true() && if()` will result in a `FunctionError` (missing arguments), but the expression `false() && if()` will not result in an error because the right operand is not evaluated.

Examples

```
eval(True && False, {"True": true, "False": false}) -> false
eval(Number && EmptyList, {"Number": 5, "EmptyList": []}) -> []
eval(foo[?a == `1` && b == `2`],
      {"foo": [{"a": 1, "b": 2}, {"a": 1, "b": 3}]}) -> [{"a": 1, "b": 2}]
```

7.6. Unary Operators

7.6.1. Not Operator

A unary NOT operator (`!`) is a boolean operator that negates the result of an expression. If the expression results in a truth-like value, NOT operator will change this value to `false`. If the expression results in a false-like value, a NOT operator will change the value to `true`.

Examples

```
eval(!True, {"True": true}) -> false
eval(!False, {"False": false}) -> true
eval(!Number, {"Number": 5}) -> false
eval(!EmptyList, {"EmptyList": []}) -> true
```

7.6.2. Minus Operator

A unary Minus operator (-) is a numeric operator that negates the value of an operand.

Examples

```
eval(-11, {}) -> -11
eval(-n, {"n": 5, "nn": -10}) -> -5
eval(-nn, {"n": 5, "nn": -10}) -> 10
eval(--n, {"n": 5, "nn": -10}) -> 5
```

8. Expressions

8.1. Chained Expressions

```
expression: expression '.' chainedExpression

chainedExpression
: identifier
| arrayExpression
| objectExpression
| functionExpression
| wildcard
;

wildcard : '*' ;
```

A chained expression is a combination of two expressions separated by the dot (.) char. A chained expression is evaluated as follows:

- Evaluate the expression on the left against the source JSON document
- Evaluate the expression on the right against the result of the left expression evaluation

In pseudo-code

```
left-evaluation = eval(left-expression, original-json-document)
result = eval(right-expression, left-evaluation)
```

A chained expression is itself an expression, so there can be multiple levels of chained expressions: **grandparent.parent.child**.

Examples

Given a JSON document: `{"foo": {"bar": "baz"}}`, and a json-formula expression: `foo.bar`, the evaluation process would be

```
left-evaluation = eval(foo, {"foo": {"bar": "baz"}}) -> {"bar": "baz"}
result = eval(bar, {"bar": "baz"}) -> "baz"
```

The final result in this example is `"baz"`.

Additional examples

```
eval(foo.bar, {"foo": {"bar": "value"}}) -> "value"
eval(foo.'bar', {"foo": {"bar": "value"}}) -> "value"
eval(foo.bar, {"foo": {"baz": "value"}}) -> null
eval(foo.bar.baz, {"foo": {"bar": {"baz": "value"}}}) -> "value"
```

8.2. Bracket Expressions

```
expression: expression bracketExpression
```

```
bracketExpression
: '[' '*' ']'
| '[' '?' expression ']'
| '[' signedInt ']'
| '[' slice ']'
| '[' ']'
;
```

```
signedInt
: '-'? INT+
;
```

From the `bracketExpression` construction, the bracketed contents provide access to the elements in an array.

- The wildcard: (`'[' '*' ']'`) variation is discussed in the [Wildcard Expressions](#) section
- The filtering: (`'[' ? expression ']'`) variation is discussed in the [Filter Expressions](#) section

8.2.1. Index Expressions

When brackets enclose a signed integer (`'[' signedInt ']'`), the integer value is used to index into an array. Indexing is 0 based where an index of 0 refers to the first element of the array. A negative

index indicates that indexing is relative to the end of the array, specifically:

```
negative-index == (length of array) + negative-index
```

Given an array of length **N**, an index of **-1** would be equal to a positive index of **N - 1**, which is the last element of the array. An index value is outside the bounds of the array when the value is greater than or equal to the length of the array or less than the negative length of the array. If an index is outside the bounds of the array then a value of **null** is returned.

Examples

```
eval(a[1], {a: [5,6,7,8,9]}) -> 6
eval(a[-2], {a: [5,6,7,8,9]}) -> 8
eval([0], ["first", "second", "third"]) -> "first"
eval([-1], ["first", "second", "third"]) -> "third"
eval([100], ["first", "second", "third"]) -> null
eval(foo[0], {"foo": ["first", "second", "third"]}) -> "first"
eval(foo[100], {"foo": ["first", "second", "third"]}) -> null
eval(foo[0][0], {"foo": [[0, 1], [1, 2]]}) -> 0
```

8.2.2. Slices

```
slice : start=signedInt? ':' stop=signedInt? (':' step=signedInt?)? ;
```

A slice expression allows you to select a contiguous subset of an array. A slice has a **start**, **stop**, and **step** value. The general form of a slice is **[start:stop:step]**. Each component of the slice is optional and can be omitted, but there must be at least one colon (:) character.

Slices in json-formula have the same semantics as python slices. If you're familiar with python slices, you're familiar with json-formula slices.

Given a **start**, **stop**, and **step** value, the sub elements in an array are extracted as follows:

- The first element in the extracted array is the index denoted by **start**.
- The last element in the extracted array is the index denoted by **stop - 1**.
- The **step** value determines the amount by which the index increases or decreases. The default step value is 1. For example, a step value of 2 will return every second value from the array. If step is negative, slicing is performed in reverse—from the last (stop) element to the start.

Slice expressions adhere to the following rules:

- If a negative start position is given, it is calculated as the total length of the array plus the given start position.
- If no start position is given, it is assumed to be 0 if the given step is greater than 0 or the end of the array if the given step is less than 0.
- If a negative stop position is given, it is calculated as the total length of the array plus the given stop position.
- If no stop position is given and the given step is greater than 0 then the stop is assumed to be the length of the array.
- If no stop position is given and the given step is less than 0, then the stop is assumed to be negative (length+1).
- If the given step is omitted, it is assumed to be 1.
- If the given step is 0, an **EvaluationError** error must be raised.
- If the element being sliced is not an array, the result is **null**.
- If the element being sliced is an array and yields no results, the result must be an empty array.

Examples

```
eval([0:4:1], [0, 1, 2, 3]) -> [0, 1, 2, 3]
eval([0:4], [0, 1, 2, 3]) -> [0, 1, 2, 3]
eval([0:3], [0, 1, 2, 3]) -> [0, 1, 2]
eval([:2], [0, 1, 2, 3]) -> [0, 1]
eval([::2], [0, 1, 2, 3]) -> [0, 2]
eval([::-1], [0, 1, 2, 3]) -> [3, 2, 1, 0]
eval([-2:], [0, 1, 2, 3]) -> [2, 3]
```

8.2.3. Flatten Operator

When the character sequence **[]** is provided as a bracket specifier, then a flattening operation occurs on the current result. The flattening operator will merge one level of sub-arrays in the current result into a single array. The flattening operator follows these processing steps:

- Create an empty result array
- Iterate over the elements of the current result
- If the current element is not an array, add to the end of the result array
- If the current element is an array, add each element of the current element to the end of the result array
- The result array is returned as a **projection**

Once the flattening operation has been performed, subsequent operations are projected onto the flattened array.

A bracketed wildcard (`[*]`) and `flatten ([[]])` behave similarly in that they produce a projection from an array. The only difference is that a bracketed wildcard preserves the original array structure while `flatten` collapses one level of array structure.

Examples

```
eval(foo[], {"foo": [[0, 1], [1, 2], 3]}) -> [0,1,1,2,3]
eval(foo[], {"foo": [[0, 1], [1, 2], [3,[4,5]]]}) -> [0,1,1,2,3,[4,5]]
eval(foo[[]], {"foo": [[0, 1], [1, 2], [3,[4,5]]]}) -> [0,1,1,2,3,4,5]
```

8.3. Projections

Projections allow you to apply an expression to a collection of elements. Projections are created when any form of a [Bracket Expression](#) transforms a source array or when a [wildcard](#) is applied to an object:

Given the source JSON:

```
{
  "items": [
    {
      "desc": "pens",
      "price": 3.23
    },
    {
      "desc": "pencils",
      "price": 1.34
    },
    {
      "desc": "staplers",
      "price": 10.79
    }
  ]
}
```

These expressions will create a projection:

- `items[*]`
- `items[]`
- `items[0:2]`
- `items[?expr]`

- `items[0].*`

When a [chained expression](#) or [bracket expression](#) is applied to a projection, their behavior is changed so that the expression is applied to each element of the projection, rather than the underlying array itself.

Examples

```
eval(items[*].desc, items) -> ["pens", "pencils", "staplers"]
eval(items[*].desc.upper(@), items) -> ["PENS", "PENCILS", "STAPLERS"]
eval(items[0].*, items) -> [
    ["pens", 3.23],
    ["pencils", 1.34],
    ["staplers", 10.79]
]
eval(items[0:2].price * 2, items) -> [6.46, 2.68]
eval(items[?price < 3], items) -> [{"desc": "pencils", "price": 1.34}]
```

A [pipe expression](#) will stop the current projection and process it as a normal array.

For example, if you wanted to sum the `price` values, this expression: `items[*].price.sum(@)` will sum each individual price, returning: `[3.23, 1.34, 10.79]`. Whereas using a pipe operator will sum the array: `items[*].price | sum(@) → 15.36`.

8.4. Paren Expressions

```
parenExpression = '(' expression ')'
```

A [parenExpression](#) allows a user to override the precedence order of an expression

e.g. `(a || b) && c`

Examples

```
eval(foo[?(a == 1 || b == 2) && c == 5],
    {"foo": [{"a": 1, "b": 2, "c": 3}, {"a": 3, "b": 4}]} -> []
```

8.5. Array Expression

```
arrayExpression : '[' expression (',' expression)* ']' ;
```


An array expression is used to extract a subset of elements from the JSON document into an array. Within the start and closing brackets are one or more expressions separated by a comma. Each expression will be evaluated against the JSON document, and each result will be added to the array

An **arrayExpression** with **N** expressions will result in an array of length **N**. Given an array expression `[expr-1,expr-2,...,expr-n]`, the evaluated expression will return `[evaluate(expr-1), evaluate(expr-2), ..., evaluate(expr-n)]`.

Given an array expression: `[n, "doubled", n * 2]` and the data `{"n": 4}`, the expression is evaluated as follows:

1. An empty array is created: `[]`
2. The expression `n` is evaluated against the source document and the result (`4`) is appended to the array
3. The literal expression `"doubled"` is appended to the array
4. The expression `n * 2` is evaluated against the source document and the result (`8`) is appended to the array

The final result will be: `[4, "doubled", 8]`.

Examples

```
eval([foo,bar], {"foo": "a", "bar": "b", "baz": "c"}) -> ["a", "b"]
eval([foo,bar[0]], {"foo": "a", "bar": ["b"], "baz": "c"}) -> ["a", "b"]
eval([foo,bar.baz], {"foo": "a", "bar": {"baz": "b"}}) -> ["a", "b"]
eval([foo,baz], {"foo": "a", "bar": "b"}) -> ["a", null]
```

The grammar contains one ambiguity: a bracket with a single signed digit e.g., `[0]` can be interpreted as a flatten operation or an **arrayExpression** with the number zero. To resolve this ambiguity, the grammar sets a precedence order so that `[-?[0-9]]` is treated as a an **index expression**. To explicitly express an array with one element, use a JSON literal: ``[0]``

8.6. Object Expression

```
objectExpression = "{" ( keyvalExpr ( "," keyvalExpr )*)? "}"
keyvalExpr = identifier ":" expression
```

An object expression is used to extract a subset of elements from the JSON document into an object. An **objectExpression** requires key names to be provided, as specified in the **keyvalExpr** rule. Given the following rule

```
keyvalExpr = identifier ":" expression
```

The **identifier** is used as the key name and the result of evaluating the **expression** is the value associated with the **identifier** key.

Each **keyvalExpr** within the **objectExpression** will correspond to a single key value pair in the created object. If a key is specified more than once, the last value will be used. Consistent with an **arrayExpression**, an **objectExpression** may not be empty. To create an empty object, use a JSON literal: `{}` .

Examples

Given an object expression `{foo: one.two, bar: bar}` and the data `{"bar": "bar", "one": {"two": "one-two"}}`, the expression is evaluated as follows:

1. An object is created: `{}`
2. A key **foo** is created whose value is the result of evaluating **one.two** against the provided JSON document: `{"foo": evaluate(one.two, <data>)}`
3. A key **bar** is created whose value is the result of evaluating the expression **bar** against the provided JSON document. If key **bar** already exists, it is replaced.

The final result will be: `{"foo": "one-two", "bar": "bar"}`.

Additional examples:

```
eval({foo: foo, bar: bar}, {"foo": "a", "bar": "b", "baz": "c"})
  -> {"foo": "a", "bar": "b"}
eval({foo: foo, firstbar: bar[0]}, {"foo": "a", "bar": ["b"]})
  -> {"foo": "a", "firstbar": "b"}
eval({foo: foo, 'bar.baz': bar.baz}, {"foo": "a", "bar": {"baz": "b"}})
  -> {"foo": "a", "bar.baz": "b"}
eval({foo: foo, baz: baz}, {"foo": "a", "bar": "b"})
  -> {"foo": "a", "baz": null}
eval({foo: foo, foo: 42}, {"foo": "a", "bar": "b"})
  -> {"foo": 42}
```

8.7. Wildcard Expressions

There are two forms of wildcard expression:

1. `[*]` from the **bracketExpression** construction:

```

bracketExpression
: '[' '*' ']'
| '[' slice ']'
| '[' ']'
| '['? expression ']'
| '[' expression ']'
;

```

2. `.*` from the `chainedExpression` construction:

```

expression : expression '.' chainedExpression

chainedExpression
: identifier
| arrayExpression
| objectExpression
| functionExpression
| wildcard
;

wildcard : '*' ;

```

The `[*]` syntax (referred to as an array wildcard expression) may be applied to arrays, and will return a [projection](#) with all the elements of the source array. If an array wildcard expression is applied to any other JSON type, a value of `null` is returned.

The `.*` syntax (referred to as an object wildcard expression) may be applied to objects and will return an array of the values from the object key/value pairs. If an object wildcard expression is applied to any other JSON type, a value of `null` is returned.

Note that JSON objects are explicitly defined as unordered. Therefore an object wildcard expression can return the values associated with the object in any order.

Examples

```

eval([*].foo, [{"foo": 1}, {"foo": 2}, {"foo": 3}]) -> [1, 2, 3]
eval([*].foo, [{"foo": 1}, {"foo": 2}, {"bar": 3}]) -> [1, 2, null]
eval(*.foo, {"a": {"foo": 1}, "b": {"foo": 2}, "c": {"bar": 1}}) ->
[1, 2, null]

```

8.8. currentNode

```

currentNode : '@' ;

```

The `currentNode` token represents the node being evaluated in the current context. The `currentNode` token is commonly used for:

- Functions that require the current node as an argument
- [Filter Expressions](#) that examine elements of an array
- Access to the current context in [projections](#).

json-formula assumes that all expressions operate on the current node. Because of this, an expression such as `@.name` would be equivalent to just `name`.

8.8.1. currentNode state

At the start of an expression, the value of the current node is the data being evaluated by the json-formula expression. As an expression is evaluated, `currentNode` must change to reflect the node being evaluated. When in a [projection](#), the current node value must be changed to the node being evaluated by the projection.

Examples

```
Given:
{
  "family": [
    {"name": "frank", "age": 22},
    {"name": "jane", "age": 23}
  ]
}

eval(@.family[0].name, {...}) -> "frank"

eval(family[].[left(@.name), age], {...}) ->
  [{"f", 22}, {"j", 23}]

eval(family[?@.age == 23], {...}) -> [{"name": "jane", "age": 23}]

eval(family[?age == 23], {...}) -> [{"name": "jane", "age": 23}]

eval(family[].name.proper(@), {...}) -> ["Frank", "Jane"]

eval(family[].age | avg(@), {...}) -> 22.5
```

8.9. Filter Expressions

```

bracketExpression
: '[' '*' ']'
| '[' slice ']'
| '[' ']'
| '[' '?' expression ']'
| '[' expression ']'
;

```

A filter expression is defined by a **bracketExpression** where the bracket contents are prefixed with a question mark character (?). A filter expression provides a way to select array elements based on a comparison to another expression. A filter expression is evaluated as follows:

- For each element in an array evaluate the **expression** against the element.
- If the expression evaluates to a truth-like value, the item (in its entirety) is added to the result array.
- If the expression does not evaluate to a truth-like value it is excluded from the result array.

A filter expression may be applied to arrays. Attempting to evaluate a filter expression against any other type will return **null**.

Examples

```

eval(
  foo[?a < b],
  {
    "foo": [
      {"a": "char", "b": "bar"},
      {"a": 2, "b": 1},
      {"a": 1, "b": 2},
      {"a": false, "b": "1"},
      {"a": 10, "b": "12"}
    ]
  })
->
[ {"a": 1, "b": 2},
  {"a": false, "b": "1"},
  {"a": 10, "b": "12"} ]

```

The five elements in the foo array are evaluated against **a < b**:

- The first element resolves to the comparison **"char" < "bar"**, and because these types are string, the comparison of code points returns **false**, and the first element is excluded from the result array.
- The second element resolves to **2 < 1**, which is **false**, so the second element is excluded from

the result array.

- The third expression resolves to `1 < 2` which evaluates to `true`, so the third element is included in the result array.
- The fourth expression resolves to `false < "1"`. Since the left hand operand is boolean, both operands are coerced to numbers and evaluated as: `0 < 1` and so the fourth element included in the result array.
- The final expression resolves to `10 < "12"`. Since we have mixed operands, the operands are coerced to numbers and evaluated as: `10 < 12` and the last element is included in the result array.

Examples

```
eval(foo[?bar==10], {"foo": [{"bar": 1}, {"bar": 10}]}) -> [{"bar": 10}]
eval([?bar==10], [{"bar": 1}, {"bar": 10}]) -> [{"bar": 10}]
eval(foo[?a=b], {"foo": [{"a": 1, "b": 2}, {"a": 2, "b": 2}]})
  -> [{"a": 2, "b": 2}]
```

8.10. Pipe Expressions

```
pipeExpression = expression '|' expression
```

A pipe expression combines two expressions, separated by the pipe (`|`) character. It is similar to a [chained expression](#) with two distinctions:

1. Any expression can be used on the right hand side. A chained expression restricts the type of expression that can be used on the right hand side.
2. A pipe expression **stops projections on the left hand side from propagating to the right hand side**. If the left expression creates a projection, the right hand side will receive the array underlying the projection.

For example, given the following data

```
{"foo": [{"bar": ["first1", "second1"]}, {"bar": ["first2", "second2"]}]}
```

The expression `foo[*].bar` gives the result of

```
[
  [
    "first1",
    "second1"
  ],
  [
    "first2",
    "second2"
  ]
]
```

The first part of the expression, `foo[*]`, creates a projection. At this point, the remaining expression, `bar` is projected onto each element of the array created from `foo[*]`. If you project the `[0]` expression, you will get the first element from each sub array. The expression `foo[*].bar[0]` will return

```
["first1", "first2"]
```

If you instead wanted **only** the first sub array, `["first1", "second1"]`, you can use a pipe expression.

```
foo[*].bar[0] -> ["first1", "first2"]
foo[*].bar | [0] -> ["first1", "second1"]
```

Examples

```
eval(foo | bar, {"foo": {"bar": "baz"}}) -> "baz"
eval(foo[*].bar | [0], {
  "foo": [{"bar": ["first1", "second1"]},
          {"bar": ["first2", "second2"]}]) -> ["first1", "second1"]
eval(foo | [0], {"foo": [0, 1, 2]}) -> 0
```

9. Functions

```
functionExpression
: NAME '(' functionArg (',' functionArg)* ')'
| NAME '(' ')'
;

functionArg
: expression
| expressionType
;

expressionType : '&' expression ;
```

json-formula has a robust set of built-in [functions](#). Each function has a signature that defines the expected types of the input and the type of the returned output.

```
return_type function_name(type $argname)
return_type function_name2(type1|type2 $argname)
```

9.1. Function parameters

Functions support the set of standard json-formula [data types](#). If the parameters cannot be coerced to match the types specified in the signature, a **TypeError** error occurs.

As a shorthand, the type **any** is used to indicate that the function argument can be any of (**array|object|number|string|boolean|null**).

The expression type, (denoted by **&expression**), is used to specify an expression that is not immediately evaluated. Instead, a reference to that expression is provided to the function. The function can then apply the expression reference as needed. It is semantically similar to an [anonymous function](#). See the [sortBy\(\)](#) function for an example of the expression type.

Function parameters can either have a specific arity or be variadic with a minimum number of arguments. If a **functionExpression** is encountered where the arity does not match, or the minimum number of arguments for a variadic function is not provided, or too many arguments are provided, then a **FunctionError** error is raised.

The result of the **functionExpression** is the result returned by the function call. If a **functionExpression** is evaluated for a function that does not exist, a **FunctionError** error is raised.

9.1.1. Array Parameters

Many functions that process scalar values also allow for the processing of arrays of values. For

example, the `round()` function may be called to process a single value: `round(1.2345, 2)` or to process an array of values: `round([1.2345, 2.3456], 2)`. The first call will return a single value, the second call will return an array of values. When processing arrays of values, and where there is more than one parameter, each parameter is converted to an array so that the function processes each value in the set of arrays. From our example above, the call to `round([1.2345, 2.3456], 2)` would be processed as if it were `round([1.2345, 2.3456], [2, 2])`, and the result would be the same as: `[round(1.2345, 2), round(2.3456, 2)]`.

Functions that accept array parameters will also accept nested arrays. Aggregating functions (`min()`, `max()`, `avg()`, `sum()`, etc.) will flatten nested arrays. e.g. `avg([2.1, 3.1, [4.1, 5.1]])` will be processed as `avg([2.1, 3.1, 4.1, 5.1])` and return 3.6.

Non-aggregating functions will return the same array hierarchy. e.g.

`upper(["a", ["b"]]) ⇒ ["A", ["B"]]` `round([2.12, 3.12, [4.12, 5.12]], 1)` will be processed as `round([2.12, 3.12, [4.12, 5.12]], [1, 1, [1, 1]])` and return `[2.1, 3.1, [4.1, 5.1]]`

These array balancing rules apply when any parameter is an array:

- All parameters will be treated as arrays
- Any scalar parameters will be converted to an array by repeating the scalar value to the length of the longest array
- All array parameters will be padded to the length of the longest array by adding null values
- The function will return an array which is the result of iterating over the elements of the arrays and applying the function logic on the values at the same index.

With nested arrays:

- Nested arrays will be flattened for aggregating functions
- Non-aggregating functions will preserve the array hierarchy and will apply the balancing rules to each element of the nested arrays

9.2. Function evaluation

Functions are evaluated in applicative order: - Each argument must be an expression - Each argument expression must be evaluated before evaluating the function - Each argument expression result must be coerced to the expected type - If coercion is not possible, a `TypeError` error is raised - The function is then called with the evaluated function arguments.

The one exception to this rule is the `if(expr, result1, result2)` function. In this case either

the `result1` expression or the `result2` expression is evaluated, depending on the outcome of `expr`.

Consider this example using the `abs()` function. Given:

```
{"foo": -1, "bar": "2"}
```

Evaluating `abs(foo)` works as follows:

1. Evaluate the input argument against the current data:

```
eval(foo, {"foo": -1, "bar": "2"}) -> -1
```

2. Coerce the type of the resolved argument if needed. In this case `-1` is of type `number` so no coercion is needed.
3. Validate the type of the coerced argument. In this case `-1` is of type `number` so it passes the type check.
4. Call the function with the resolved argument:

```
abs(-1) -> 1
```

Below is the same steps for evaluating `abs(bar)`:

1. Evaluate the input argument against the current data:

```
eval(bar, {"foo": -1, "bar": "2"}) -> "2"
```

2. Attempt to coerce the result to the required number type. In this case, coerce `"2"` to `2`.
3. Validate the type of the coerced argument. In this case `2` is of type `number` so it passes the type check.
4. Call the function with the resolved and coerced argument:

```
abs(2) -> 2
```

Function expressions are allowed as the child element of a `chained expression`. The function is then evaluated in the context of the parent expression result. For example: `[1,2,3].sum(@) -> 6`.

When the parent expression is a `projection` a chained function expression will be applied to every

element in the projection. For example, given the input data of `["1", "2", "3", "notanumber", true]`, the following expression can be used to convert all elements to numbers:

```
eval([], toNumber(@), ["1", "2", "3", "notanumber", null, true]) ->
[1,2,3,null,0,1]
```

9.3. Function Reference

`abs` `acos` `and` `asin` `atan2` `avg` `avgA` `casefold` `ceil` `codePoint` `contains` `cos` `datedif` `datetime` `day` `debug` `deepScan` `endsWith` `entries` `eomonth` `exp` `false` `find` `floor` `fromCodePoint` `fromEntries` `fround` `hasProperty` `hour` `if` `join` `keys` `left` `length` `log` `log10` `lower` `map` `max` `maxA` `merge` `mid` `millisecond` `min` `minA` `minute` `mod` `month` `not` `notNull` `now` `null` `or` `power` `proper` `random` `reduce` `register` `registerWithParams` `replace` `rept` `reverse` `right` `round` `search` `second` `sign` `sin` `sort` `sortBy` `split` `sqrt` `startsWith` `stdev` `stdevA` `stdevp` `stdevp` `substitute` `sum` `tan` `time` `toArray` `toDate` `today` `toNumber` `toString` `trim` `true` `trunc` `type` `unique` `upper` `value` `values` `weekday` `year` `zip`

9.3.1. abs

abs(value) ⇒ number | Array.<number>

Description

Find the absolute (non-negative) value of the provided argument `value`.

Returns

number | Array.<number> - If `value < 0`, returns `-value`, otherwise returns `value`

Parameters

Param	Type	Description
value	number Array.<number>	A numeric value

Example

```
abs(-1) // returns 1
```

9.3.2. acos

acos(cosine) ⇒ number | Array.<number>

Description

Compute the inverse cosine (in radians) of a number.

Returns

number | Array.<number> - The inverse cosine angle in radians between 0 and PI

Parameters

Param	Type	Description
cosine	number Array.<number>	A number between -1 and 1, inclusive, representing the angle's cosine value.

Example

```
acos(0) => 1.5707963267948966
```

9.3.3. and

and(firstOperand, [...additionalOperands]) ⇒ boolean

Description

Finds the logical AND result of all parameters. If the parameters are not boolean they will be [cast to boolean](#). Note the related [And Operator](#).

Returns

boolean - The logical result of applying AND to all parameters

Parameters

Param	Type	Description
firstOperand	any	logical expression
[...additionalOperands]	any	any number of additional expressions

Example

```
and(10 > 8, length("foo") < 5) // returns true
and(`null`, length("foo") < 5) // returns false
```

9.3.4. asin

asin(sine) ⇒ number | Array.<number>

Description

Compute the inverse sine (in radians) of a number.

Returns

number | Array.<number> - The inverse sine angle in radians between $-\pi/2$ and $\pi/2$

Parameters

Param	Type	Description
sine	number Array.<number>	A number between -1 and 1, inclusive, representing the angle's sine value.

Example

```
Math.asin(0) => 0
```

9.3.5. atan2

atan2(y, x) ⇒ number | Array.<number>

Description

Compute the angle in the plane (in radians) between the positive x-axis and the ray from (0, 0) to the point (x, y)

Returns

number | Array.<number> - The angle in radians (between $-\pi$ and π), between the positive x-axis and the ray from (0, 0) to the point (x, y).

Parameters

Param	Type	Description
y	number Array.<number>	The y coordinate of the point
x	number Array.<number>	The x coordinate of the point

Example

```
atan2(20,10) => 1.1071487177940904
```

9.3.6. avg

avg(elements) ⇒ number

Description

Finds the average of the elements in an array. Non-numeric values (text, boolean, null, object) are ignored. If there are nested arrays, they are flattened. If the array is empty, an evaluation error is thrown

Returns

number - average value

Parameters

Param	Type	Description
elements	Array.<any>	array of values

Example

```
avg([1, 2, 3]) // returns 2
```

9.3.7. avgA

avgA(elements) ⇒ number

Description

Finds the average of the elements in an array, converting strings and booleans to number. If any conversions to number fail, a type error is thrown. If there are nested arrays, they are flattened. If the array is empty, an evaluation error is thrown

Returns

number - average value

Parameters

Param	Type	Description
elements	Array.<number>	array of numeric values

Example

```
avgA([1, 2, "3", null()]) // returns 2
```

9.3.8. casefold

casefold(input) ⇒ string | Array.<string>

Description

Generates a lower-case string of the **input** string using locale-specific mappings. e.g. Strings with German letter ß (eszett) can be compared to “ss”

Returns

string | Array.<string> - A new string converted to lower case

Parameters

Param	Type	Description
input	string Array.<string>	string to casefold

Example

```
casefold("AbC") // returns "abc"
```

9.3.9. ceil

ceil(num) ⇒ integer | Array.<integer>

Description

Finds the next highest integer value of the argument **num** by rounding up if necessary. i.e. ceil() rounds toward positive infinity.

Returns

integer | Array.<integer> - The smallest integer greater than or equal to num

Parameters

Param	Type	Description
num	number Array.<number>	numeric value

Example

```
ceil(10) // returns 10
```

```
ceil(10.4) // return 11
```

9.3.10. codePoint

codePoint(str) ⇒ integer | Array.<integer>

Description

Retrieve the first code point from a string

Returns

integer | Array.<integer> - Unicode code point value. If the input string is empty, returns **null**.

Parameters

Param	Type	Description
str	string Array.<string>	source string.

Example

```
codePoint("ABC") // 65
```

9.3.11. contains

contains(subject, search) ⇒ boolean

Description

Determines if the given **subject** contains **search**. If **subject** is an array, this function returns true if one of the elements in the array is equal to the **search** value. If **subject** is a string, return true if the string contains the **search** value.

Returns

boolean - true if found

Parameters

Param	Type	Description
subject	array string	The element to be searched

Param	Type	Description
search	any	element to find. If subject is an array, search for an exact match for search in the array. If subject is a string, search must also be a string.

Example

```
contains([1, 2, 3, 4], 2) // returns true
contains([1, 2, 3, 4], -1) // returns false
contains("Abcd", "d") // returns true
contains("Abcd", "x") // returns false
```

9.3.12. cos

cos(angle) ⇒ number | Array.<number>

Description

Compute the cosine (in radians) of a number.

Returns

number | Array.<number> - The cosine of the angle, between -1 and 1, inclusive.

Parameters

Param	Type	Description
angle	number Array.<number>	A number representing an angle in radians

Example

```
cos(1.0471975512) => 0.4999999999970535
```

9.3.13. datedif

datedif(start_date, end_date, unit) ⇒ integer | Array.<integer>

Description

Return difference between two date values. The measurement of the difference is determined by

the **unit** parameter. One of:

- **y** the number of whole years between **start_date** and **end_date**
- **m** the number of whole months between **start_date** and **end_date**.
- **d** the number of days between **start_date** and **end_date**
- **ym** the number of whole months between **start_date** and **end_date** after subtracting whole years.
- **yd** the number of days between **start_date** and **end_date**, assuming **start_date** and **end_date** were no more than one year apart

Returns

integer | Array.<integer> - The number of days/months/years difference

Parameters

Param	Type	Description
start_date	number Array.<number>	The starting date/time value . Date/time values can be generated using the datetime , toDate , today , now and time functions.
end_date	number Array.<number>	The end date/time value – must be greater or equal to start_date. If not, an error will be thrown.
unit	string Array.<string>	Case-insensitive string representing the unit of time to measure. An unrecognized unit will result in an error.

Example

```
datedif(datetime(2001, 1, 1), datetime(2003, 1, 1), "y") // returns 2
datedif(datetime(2001, 6, 1), datetime(2003, 8, 15), "D") // returns 805
// 805 days between June 1, 2001, and August 15, 2003
datedif(datetime(2001, 6, 1), datetime(2003, 8, 15), "YD") // returns 75
// 75 days between June 1 and August 15, ignoring the years of the dates (75)
```

9.3.14. datetime

datetime(year, month, day, [hours], [minutes], [seconds], [milliseconds]) ⇒ number

Description

Generate a date/time value from individual date/time parts. If any of the units are greater than their normal range, the overflow will be added to the next greater unit. e.g. specifying 25 hours will increment the day value by 1. Similarly, negative values will decrement the next greater unit. e.g. datetime(year, month, day - 30) will return a date 30 days earlier.

Returns

number - A [date/time value](#) to be used with other date/time functions

Parameters

Param	Type	Default	Description
year	integer		The year to use for date construction. Values from 0 to 99 map to the years 1900 to 1999. All other values are the actual year
month	integer		The month: beginning with 1 for January to 12 for December.
day	integer		The day of the month.
[hours]	integer	0	Integer value between 0 and 23 representing the hour of the day.
[minutes]	integer	0	Integer value representing the minute segment of a time.
[seconds]	integer	0	Integer value representing the second segment of a time.

Param	Type	Default	Description
[milliseconds]	integer	0	Integer value representing the millisecond segment of a time.

Example

```
datetime(2010, 10, 10) // returns representation of October 10, 2010
datetime(2010, 2, 28) // returns representation of February 28, 2010
datetime(2023,13,5) | year(@) & "/" & month(@) // returns 2024/1
```

9.3.15. day

day(date) ⇒ integer | Array.<integer>

Description

Finds the day of the month for a date value

Returns

integer | Array.<integer> - The day of the month ranging from 1 to 31.

Parameters

Param	Type	Description
date	number Array.<number>	date/time value generated using the datetime , toDate , today , now and time functions.

Example

```
day(datetime(2008,5,23)) // returns 23
```

9.3.16. debug

debug(arg, [displayValue]) ⇒ any

Description

Debug a json-formula expression. The `debug()` function allows users to inspect a sub-expression within a formula.

Returns

any - The value of the `arg` parameter

Parameters

Param	Type	Default	Description
<code>arg</code>	any		The expression to return from <code>debug()</code> function, and the default expression to be debugged. May be any type except an expression.
<code>[displayValue]</code>	any expression	<code>arg</code>	Optionally override the value to be debugged. <code>displayValue</code> may be an expression to be evaluated with the context of <code>arg</code> .

Example

```
avg([1,2,3] * [2,3,4]).debug(@).round(@,3) // 6.667
avg(debug([1,2,3] * [2,3,4],&"average of: " &toString(@)).round(@,3) // 6.667
```

9.3.17. deepScan

deepScan(object, name) ⇒ Array.<any>

Description

Performs a depth-first search of a nested hierarchy to return an array of key values that match a `name`. The name can be either a key into an object or an array index. This is similar to the Descendant Accessor operator (`. .`) from [E4X](#).

Returns

Array.<any> - The array of matched elements

Parameters

Param	Type	Description
object	object array null	The starting object or array where we start the search
name	string integer	The name (or index position) of the elements to find. If name is a string, search for nested objects with a matching key. If name is an integer, search for nested arrays with a matching index.

Example

```
deepScan({a : {b1 : {c : 2}, b2 : {c : 3}}}, "c") // returns [2, 3]
```

9.3.18. endsWith

endsWith(subject, suffix) ⇒ boolean | Array.<boolean>

Description

Determines if the **subject** string ends with a specific **suffix**

Returns

boolean | Array.<boolean> - true if the **suffix** value is at the end of the **subject**

Parameters

Param	Type	Description
subject	string Array.<string>	source string in which to search
suffix	string Array.<string>	search string

Example

```
endsWith("Abcd", "d") // returns true
endsWith("Abcd", "A") // returns false
```

9.3.19. entries

entries(obj) ⇒ Array.<any>

Description

Returns an array of `[key, value]` pairs from an object or array. The `fromEntries()` function may be used to convert an array to an object.

Returns

`Array.<any>` - an array of arrays where each child array has two elements representing the key and value of a pair

Parameters

Param	Type	Description
obj	object array	source object or array

Example

```
entries({a: 1, b: 2}) // returns [["a", 1], ["b", 2]]
entries([4,5]) // returns [["0", 4], ["1", 5]]
```

9.3.20. eomonth

`eomonth(startDate, monthAdd)` ⇒ `number` | `Array.<number>`

Description

Finds the date value of the end of a month, given `startDate` plus `monthAdd` months

Returns

`number` | `Array.<number>` - the date of the last day of the month

Parameters

Param	Type	Description
startDate	<code>number</code> <code>Array.<number></code>	The base date to start from. Date/time values can be generated using the datetime , toDate , today , now and time functions.
monthAdd	<code>integer</code> <code>Array.<integer></code>	Number of months to add to start date

Example

```
eomonth(datetime(2011, 1, 1), 1) | [month(@), day(@)] // returns [2, 28]
eomonth(datetime(2011, 1, 1), -3) | [month(@), day(@)] // returns [10, 31]
```

9.3.21. exp

exp(x) ⇒ number | Array.<number>

Description

Finds e (the base of natural logarithms) raised to a power. (i.e. e^x)

Returns

number | Array.<number> - e (the base of natural logarithms) raised to power x

Parameters

Param	Type	Description
x	number Array.<number>	A numeric expression representing the power of e.

Example

```
exp(10) // returns 22026.465794806718
```

9.3.22. false

false() ⇒ boolean

Description

Return constant boolean false value. Expressions may also use the JSON literal: `\ 'false\ '`

Returns

boolean - constant boolean value `false`

9.3.23. find

find(findText, withinText, [start]) ⇒ integer | null | Array.<integer>

Description

Finds and returns the index of query in text from a start position

Returns

integer | null | Array.<integer> - The position of the found string, null if not found.

Parameters

Param	Type	Default	Description
findText	string Array.<string>		string to search
withinText	string Array.<string>		text to be searched
[start]	integer Array.<integer>	0	zero-based position to start searching. If specified, start must be greater than or equal to 0

Example

```
find("m", "abm") // returns 2
find("M", "abMcdM", 3) // returns 5
find("M", "ab") // returns `null`
find("M", "abMcdM", 2) // returns 2
```

9.3.24. floor

floor(num) ⇒ integer | Array.<integer>

Description

Calculates the next lowest integer value of the argument **num** by rounding down if necessary. i.e. floor() rounds toward negative infinity.

Returns

integer | Array.<integer> - The largest integer smaller than or equal to num

Parameters

Param	Type	Description
num	number Array.<number>	numeric value

Example

```
floor(10.4) // returns 10
floor(10) // returns 10
```

9.3.25. fromCodePoint

fromCodePoint(codePoint) ⇒ string

Description

Create a string from a code point.

Returns

string - A string from the given code point(s)

Parameters

Param	Type	Description
codePoint	integer Array.<integer>	An integer or array of integers between 0 and 0x10FFFF (inclusive) representing Unicode code point(s).

Example

```
fromCodePoint(65) // "A"  
fromCodePoint(65) == "\u0041" // true
```

9.3.26. fromEntries

fromEntries(pairs) ⇒ object

Description

Returns an object by transforming a list of key-value **pairs** into an object. **fromEntries()** is the inverse operation of **entries()**. If the nested arrays are not of the form: **[key, value]** (where key is a string), an error will be thrown.

Returns

object - An object constructed from the provided key-value pairs

Parameters

Param	Type	Description
pairs	Array.<any>	A nested array of key-value pairs to create the object from The nested arrays must have exactly two values, where the first value is a string. If a key is specified more than once, the last occurrence will override any previous value.

Example

```
fromEntries([["a", 1], ["b", 2]]) // returns {a: 1, b: 2}
```

9.3.27. fround

fround(num) ⇒ number | Array.<number>

Description

Compute the nearest 32-bit single precision float representation of a number

Returns

number | Array.<number> - The rounded representation of **num**

Parameters

Param	Type	Description
num	number Array.<number>	input to be rounded

Example

```
fround(2147483650.987) => 2147483648
fround(100.444444444444444444444444444444) => 100.44444274902344
```

9.3.28. hasProperty

hasProperty(subject, name) ⇒ boolean

Description

Determine if an object has a property or if an array index is in range.

Returns

boolean - true if the element exists

Parameters

Param	Type	Description
subject	object array null	source object or array. When querying for hidden properties , subject may be any data type.
name	string integer	The name (or index position) of the element to find. if subject is an array, name must be an integer; if subject is an object, name must be a string.

Example

```
hasProperty({a: 1, b: 2}, "a") // returns true
hasProperty(["apples", "oranges"], 3) // returns false
hasProperty(`null`, "a") // returns false
```

9.3.29. hour

hour(date) ⇒ integer | Array.<integer>

Description

Extract the hour from a [date/time value](#)

Returns

integer | Array.<integer> - value between 0 and 23

Parameters

Param	Type	Description
date	number Array.<number>	The datetime/time for which the hour is to be returned. Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
hour(datetime(2008,5,23,12, 0, 0)) // returns 12
hour(time(12, 0, 0)) // returns 12
```

9.3.30. if

if(condition, result1, result2) ⇒ any

Description

Return one of two values **result1** or **result2**, depending on the **condition**

Returns

any - either result1 or result2

Parameters

Param	Type	Description
condition	any	boolean result of a logical expression
result1	any	if condition is true
result2	any	if condition is false

Example

```
if(true(), 1, 2) // returns 1
if(false(), 1, 2) // returns 2
```

9.3.31. join

join(array, glue) ⇒ string

Description

Combines all the elements from the provided array, joined together using the **glue** argument as a separator between each array element.

Returns

string - String representation of the array

Parameters

Param	Type	Description
array	Array.<any>	array of values that will be converted to strings using toString()
glue	string	

Example

```
join(["a", "b", "c"], ",") // returns "a,b,c"  
join(["apples", "bananas"], " and ") // returns "apples and bananas"  
join([1, 2, 3, null()], "|") // returns "1|2|3|null"
```

9.3.32. keys

keys(obj) ⇒ array

Description

Generates an array of the keys of the input object. If the object is null, the value return an empty array

Returns

array - the array of all the key names

Parameters

Param	Type	Description
obj	object	the object to examine

Example

```
keys({a : 3, b : 4}) // returns ["a", "b"]
```

9.3.33. left

left(subject, [elements]) ⇒ string | array

Description

Return a substring from the start of a string or the left-most elements of an array

Parameters

Param	Type	Default	Description
subject	string array		The source text/array of code points/elements
[elements]	integer	1	number of elements to pick

Example

```
left("Sale Price", 4) // returns "Sale"
left("Sweden") // returns "S"
left([4, 5, 6], 2) // returns [4, 5]
```

9.3.34. length

length(subject) ⇒ integer

Description

Calculates the length of the input argument based on types:

- string: returns the number of unicode code points
- array: returns the number of array elements
- object: returns the number of key-value pairs

Returns

integer - the length of the input subject

Parameters

Param	Type	Description
subject	string array object	subject whose length to calculate

Example

```
length(`[]`) // returns 0
length("") // returns 0
length("abcd") // returns 4
length([1, 2, 3, 4]) // returns 4
length(`{}`) // returns 0
length({a : 3, b : 4}) // returns 2
```

9.3.35. log

log(num) ⇒ number | Array.<number>

Description

Compute the natural logarithm (base e) of a number

Returns

number | Array.<number> - The natural log value

Parameters

Param	Type	Description
num	number Array.<number>	A number greater than zero

Example

```
log(10) // 2.302585092994046
```

9.3.36. log10

log10(num) ⇒ number | Array.<number>

Description

Compute the base 10 logarithm of a number.

Returns

number | Array.<number> - The base 10 log result

Parameters

Param	Type	Description
num	number Array.<number>	A number greater than or equal to zero

Example

```
log10(100000) // 5
```

9.3.37. lower

lower(input) ⇒ string | Array.<string>

Description

Converts all the alphabetic code points in a string to lowercase.

Returns

string | Array.<string> - the lower case value of the input string

Parameters

Param	Type	Description
input	string Array.<string>	input string

Example

```
lower("E. E. Cummings") // returns "e. e. cummings"
```

9.3.38. map

map(elements, expr) ⇒ array

Description

Apply an expression to every element in an array and return the array of results. An input array of length N will return an array of length N.

Returns

array - the mapped array

Parameters

Param	Type	Description
elements	array	array of elements to process
expr	expression	expression to evaluate

Example

```
map([1, 2, 3, 4], &(@ + 1)) // returns [2, 3, 4, 5]
map(["doe", "nick", "chris"], &length(@)) // returns [3, 4, 5]
```

9.3.39. max

max(...collection) ⇒ number

Description

Calculates the largest value in the input numbers. Any values that are not numbers (null, boolean, strings, objects) will be ignored. If any parameters are arrays, the arrays will be flattened. If no numbers are provided, the function will return zero.

Returns

number - the largest value found

Parameters

Param	Type	Description
...collection	array any	values/array(s) in which the maximum element is to be calculated

Example

```
max([1, 2, 3], [4, 5, 6]) // returns 6
max(["a", "a1", "b"], null(), true()) // returns 0
max(8, 10, 12, "14") // returns 12
```

9.3.40. maxA

maxA(...collection) ⇒ number

Description

Calculates the largest value in the input values, coercing parameters to numbers. Null values are

ignored. If any parameters cannot be converted to a number, the function will fail with a type error. If any parameters are arrays, the arrays will be flattened. If no numbers are provided, the function will return zero.

Returns

number - the largest value found

Parameters

Param	Type	Description
...collection	any	values/array(s) in which the maximum element is to be calculated

Example

```
maxA([1, 2, 3], [4, 5, 6]) // returns 6
maxA(["a", "a1", "b", null()]) // error
maxA(8, 10, 12, "14") // returns 14
```

9.3.41. merge

merge(...args) ⇒ object

Description

Accepts one or more objects, and returns a single object with all objects merged. The first object is copied, and then each key value pair from each subsequent object are added to the first object. Duplicate keys in subsequent objects will override those found in earlier objects.

Returns

object - The combined object

Parameters

Param	Type
...args	object

Example

```
merge({a: 1, b: 2}, {c: 3, d: 4}) // returns {a: 1, b: 2, c: 3, d: 4}
merge({a: 1, b: 2}, {a: 3, d: 4}) // returns {a: 3, b: 2, d: 4}
```

9.3.42. mid

mid(subject, startPos, length) ⇒ string | array

Description

Extracts a substring from text, or a subset from an array.

Returns

string | array - The resulting substring or array subset of elements

Parameters

Param	Type	Description
subject	string array	the text string or array of elements from which to extract.
startPos	integer	the zero-based position of the first code point or element to extract.
length	integer	The number of code points or elements to return from the string or array. If greater than the length of subject the length of the subject is used.

Example

```
mid("Fluid Flow", 0, 5) // returns "Fluid"
mid("Fluid Flow", 6, 20) // returns "Flow"
mid("Fluid Flow", 20, 5) // returns ""
mid([0,1,2,3,4,5,6,7,8,9], 2, 3) // returns [2,3,4]
```

9.3.43. millisecond

millisecond(date) ⇒ integer | Array.<integer>

Description

Extract the milliseconds of the time value in a [date/time value](#).

Returns

integer | Array.<integer> - The number of milliseconds: 0 through 999

Parameters

Param	Type	Description
date	number Array.<number>	datetime/time for which the millisecond is to be returned. Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
millisecond(datetime(2008, 5, 23, 12, 10, 53, 42)) // returns 42
```

9.3.44. min

min(...collection) ⇒ number

Description

Calculates the smallest value in the input numbers. Any values that are not numbers (null, boolean, string, object) will be ignored. If any parameters are arrays, the arrays will be flattened. If no numbers are provided, the function will return zero.

Returns

number - the smallest value found

Parameters

Param	Type	Description
...collection	Array.<any> any	Values/arrays to search for the minimum value

Example

```
min([1, 2, 3], [4, 5, 6]) // returns 1  
min("4", 8, 10, 12, null()) // returns 4
```

9.3.45. minA

minA(...collection) ⇒ number

Description

Calculates the smallest value in the input values, coercing parameters to numbers. Null values are ignored. If any parameters cannot be converted to a number, the function will fail with a type error. If any parameters are arrays, the arrays will be flattened. If no numbers are provided, the function will return zero.

Returns

number - the largest value found

Parameters

Param	Type	Description
...collection	Array.<any> any	values/array(s) in which the maximum element is to be calculated

Example

```
minA([1, 2, 3], [4, 5, 6]) // returns 1
minA("4", 8, 10, 12, null()) // returns 4
```

9.3.46. minute

minute(date) ⇒ integer | Array.<integer>

Description

Extract the minute (0 through 59) from a [date/time value](#)

Returns

integer | Array.<integer> - Number of minutes in the time portion of the date/time value

Parameters

Param	Type	Description
date	number Array.<number>	A datetime/time value. Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
minute(datetime(2008,5,23,12, 10, 0)) // returns 10
minute(time(12, 10, 0)) // returns 10
```

9.3.47. mod

mod(dividend, divisor) ⇒ number | Array.<number>

Description

Return the remainder when one number is divided by another number.

Returns

number | Array.<number> - Computes the remainder of **dividend/divisor**. If **dividend** is negative, the result will also be negative. If **dividend** is zero, an error is thrown.

Parameters

Param	Type	Description
dividend	number Array.<number>	The number for which to find the remainder.
divisor	number Array.<number>	The number by which to divide number.

Example

```
mod(3, 2) // returns 1
mod(-3, 2) // returns -1
```

9.3.48. month

month(date) ⇒ integer | Array.<integer>

Description

Finds the month of a date.

Returns

integer | Array.<integer> - The month number value, ranging from 1 (January) to 12 (December)

Parameters

Param	Type	Description
date	number Array.<number>	source date/time value . Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
month(datetime(2008,5,23)) // returns 5
```

9.3.49. not

not(value) ⇒ boolean

Description

Compute logical NOT of a value. If the parameter is not boolean it will be [cast to boolean](#) Note the related [unary NOT operator](#).

Returns

boolean - The logical NOT applied to the input parameter

Parameters

Param	Type	Description
value	any	any data type

Example

```
not(length("bar") > 0) // returns false
not(false()) // returns true
not("abcd") // returns false
not("") // returns true
```

9.3.50. notNull

notNull(...argument) ⇒ any

Description

Finds the first argument that does not resolve to [null](#). This function accepts one or more

arguments, and will evaluate them in order until a non-null argument is encountered. If all arguments values resolve to null, then return a null value.

Parameters

Param	Type
...argument	any

Example

```
notNull(1, 2, 3, 4, `null`) // returns 1
notNull(`null`, 2, 3, 4, `null`) // returns 2
```

9.3.51. now

now() ⇒ number

Description

Retrieve the current date/time.

Returns

number - representation of the [current date/time value](#).

9.3.52. null

null() ⇒ boolean

Description

Return constant null value. Expressions may also use the JSON literal: `\`null\``

Returns

boolean - True

9.3.53. or

or(first, [...operand]) ⇒ boolean

Description

Determines the logical OR result of a set of parameters. If the parameters are not boolean they will be [cast to boolean](#). Note the related [Or Operator](#).

Returns

boolean - The logical result of applying OR to all parameters

Parameters

Param	Type	Description
first	any	logical expression
[...operand]	any	any number of additional expressions

Example

```
or((x / 2) == y, (y * 2) == x) // true
```

9.3.54. power

power(a, x) ⇒ number | Array.<number>

Description

Computes **a** raised to a power **x**. (a^x)

Parameters

Param	Type	Description
a	number Array.<number>	The base number – can be any real number.
x	number Array.<number>	The exponent to which the base number is raised.

Example

```
power(10, 2) // returns 100 (10 raised to power 2)
```

9.3.55. proper

proper(text) ⇒ string | Array.<string>

Description

Apply proper casing to a string. Proper casing is where the first letter of each word is converted to an uppercase letter and the rest of the letters in the word converted to lowercase. Words are

demarcated by whitespace, punctuation, or numbers. Specifically, any character(s) matching the regular expression: `[\s\d\p{P}]+`.

Returns

string | Array.<string> - source string with proper casing applied.

Parameters

Param	Type	Description
text	string Array.<string>	source string

Example

```
proper("this is a TITLE") // returns "This Is A Title"
proper("2-way street") // returns "2-Way Street"
proper("76BudGet") // returns "76Budget"
```

9.3.56. random

random() ⇒ number

Description

Generate a pseudo random number.

Returns

number - A value greater than or equal to zero, and less than one.

Example

```
random() // 0.022585461160693265
```

9.3.57. reduce

reduce(elements, expr, initialValue) ⇒ any

Description

Executes a user-supplied reducer expression on each element of an array, in order, passing in the return value from the expression from the preceding element. The final result of running the reducer across all elements of the input array is a single value. The expression can access the following properties of the current object:

- accumulated: accumulated value based on the previous expression. For the first array element

use the `initialValue` parameter. If not provided, then `null`

- `current`: current element to process
- `index`: index of the current element in the array
- `array`: original array

Parameters

Param	Type	Description
<code>elements</code>	array	array of elements on which the expression will be evaluated
<code>expr</code>	expression	reducer expression to be executed on each element
<code>initialValue</code>	any	the accumulated value to pass to the first array element

Example

```
reduce([1, 2, 3], &(accumulated + current)) // returns 6
// find maximum entry by age
reduce(
  [{age: 10, name: "Joe"}, {age: 20, name: "John"}],
  &max(@.accumulated.age, @.current.age), @[0].age)
reduce([3, 3, 3], &accumulated * current, 1) // returns 27
```

9.3.58. register

register(functionName, expr) ⇒ Object

Description

Register a function. The registered function may take one parameter. If more parameters are needed, combine them in an array or object. A function may not be re-registered with a different definition. Note that implementations are not required to provide `register()` in order to be conformant. Built-in functions may not be overridden.

Returns

Object - returns an empty object

Parameters

Param	Type	Description
functionName	string	Name of the function to register. <code>functionName</code> must begin with an underscore or uppercase letter and follow the regular expression pattern: <code>{caret}{startsb}_A-Z{endsb}{startsb}_a-zA-Z0-9\${endsb}{asterisk}\$</code>
expr	expression	Expression to execute with this function call

Example

```
register("_product", &@[0] * @[1]) // can now call: _product([2,21]) => returns 42
register("_ltrim", &split(@,"").reduce(@, &accumulated & current | if(@ = " ", "", @), ""))
// _ltrim(" abc ") => returns "abc "
```

9.3.59. registerWithParams

registerWithParams(functionName, expr) ⇒ Object

Description

Register a function that accepts multiple parameters. A function may not be re-registered with a different definition. Note that implementations are not required to provide `registerWithParams()` in order to be conformant. Built-in functions may not be overridden.

Returns

Object - returns an empty object

Parameters

Param	Type	Description
functionName	string	Name of the function to register. functionName must begin with an underscore or uppercase letter and follow the regular expression pattern: {caret}{startsb}_A-Z{endsb}{startsb}_a-zA-Z0-9\${endsb}{asterisk}\$
expr	expression	Expression to execute with this function call. Parameters are passed as an array.

Example

```
registerWithParams("Product", &@[0] * @[1])
// can now call: Product(2,21) => returns 42
registerWithParams(
  "Ltrim",
  &split(@[0], "").reduce(@, &accumulated & current | if(@ = " ", "", @), "")
)
// Ltrim("  abc ") => returns "abc "
```

9.3.60. replace

replace(subject, start, length, replacement) ⇒ string | array

Description

Generates text (or an array) where we substitute elements at a given start position and length, with new text (or array elements).

Returns

string | array - the resulting text or array

Parameters

Param	Type	Description
subject	string array	original text or array

Param	Type	Description
start	integer	zero-based index in the original text from where to begin the replacement. Must be greater than or equal to 0.
length	integer	number of code points to be replaced. If start length is greater than the length of subject , all text past start will be replaced.
replacement	any	Replacement to insert at the start index. If subject is an array, and replacement is an array, the replacement array elements will be inserted into the subject array. If subject is an array and replacement is not an array, the replacement will be inserted as a single element in subject . If subject is a string, the replacement will be coerced to a string.

Example

```
replace("abcdefghijk", 5, 5, "*") // returns abcde*k
replace("2009",2,2,"10") // returns  2010
replace("123456",0,3,"@") // returns @456
replace(["blue","black","white","red"], 1, 2, ["green"]) // returns
["blue","green","red"]
```

9.3.61. rept

rept(text, count) ⇒ string | Array.<string>

Description

Return text repeated **count** times.

Returns

string | Array.<string> - Text generated from the repeated text. if **count** is zero, returns an empty string.

Parameters

Param	Type	Description
text	string Array.<string>	text to repeat
count	integer Array.<integer>	number of times to repeat the text. Must be greater than or equal to 0.

Example

```
rept("x", 5) // returns "xxxxx"
```

9.3.62. reverse

reverse(subject) ⇒ array

Description

Reverses the order of an array or the order of code points in a string

Returns

array - The resulting reversed array or string

Parameters

Param	Type	Description
subject	string array	the source to be reversed

Example

```
reverse(["a", "b", "c"]) // returns ["c", "b", "a"]
```

9.3.63. right

right(subject, [elements]) ⇒ string | array

Description

Generates a string from the right-most code points of a string or a subset of elements from the end of an array

Returns

string | array - The extracted substring or array subset Returns null if the number of elements is less than 0

Parameters

Param	Type	Default	Description
subject	string array		The text/array containing the code points/elements to extract
[elements]	integer	1	number of elements to pick

Example

```
right("Sale Price", 4) // returns "rice"
right("Sweden") // returns "n"
right([4, 5, 6], 2) // returns [5, 6]
```

9.3.64. round

round(num, [precision]) ⇒ number | Array.<number>

Description

Round a number to a specified precision:

- If **precision** is greater than zero, round to the specified number of decimal places.
- If **precision** is 0, round to the nearest integer.
- If **precision** is less than 0, round to the left of the decimal point.

Returns

number | Array.<number> - rounded value. Rounding a half value will round up.

Parameters

Param	Type	Default	Description
num	number Array.<number>		number to round
[precision]	integer Array.<integer>	0	precision to use for the rounding operation.

Example

```
round(2.15, 1) // returns 2.2
round(626.3,-3) // returns 1000 (Rounds 626.3 to the nearest multiple of 1000)
round(626.3, 0) // returns 626
round(1.98,-1) // returns 0 (Rounds 1.98 to the nearest multiple of 10)
round(-50.55,-2) // -100 (round -50.55 to the nearest multiple of 100)
round(1.95583) // 2
round(-1.5) // -1
```

9.3.65. search

search(findText, withinText, [startPos]) ⇒ array

Description

Perform a wildcard search. The search is case-sensitive and supports two forms of wildcards: **{asterisk}** finds a sequence of code points and **?** finds a single code point. To use **{asterisk}** or **?** or **{backslash}** as text values, precede them with an escape (**{backslash}**) character. Note that the wildcard search is not greedy. e.g. **search("a{asterisk}b", "abb")** will return **[0, "ab"]** Not **[0, "abb"]**

Returns

array - returns an array with two values:

- The start position of the found text and the text string that was found.
- If a match was not found, an empty array is returned.

Parameters

Param	Type	Default	Description
findText	string Array.<string>		the search string – which may include wild cards.
withinText	string Array.<string>		The string to search.

Param	Type	Default	Description
[startPos]	integer Array.<integer>	0	The zero-based position of withinText to start searching. A negative value is not allowed.

Example

```
search("a?c", "acabc") // returns [2, "abc"]
```

9.3.66. second

second(date) ⇒ integer | Array.<integer>

Description

Extract the seconds of the time value in a [date/time value](#).

Returns

integer | Array.<integer> - The number of seconds: 0 through 59

Parameters

Param	Type	Description
date	number Array.<number>	datetime/time for which the second is to be returned. Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
second(datetime(2008,5,23,12, 10, 53)) // returns 53
second(time(12, 10, 53)) // returns 53
```

9.3.67. sign

sign(num) ⇒ number | Array.<number>

Description

Computes the sign of a number passed as argument.

Returns

number | Array.<number> - returns 1 or -1, indicating the sign of **num**. If the **num** is 0, it will return 0.

Parameters

Param	Type	Description
num	number Array.<number>	any number

Example

```
sign(5) // 1
sign(-5) // -1
sign(0) // 0
```

9.3.68. sin

sin(angle) ⇒ number | Array.<number>

Description

Computes the sine of a number in radians

Returns

number | Array.<number> - The sine of **angle**, between -1 and 1, inclusive

Parameters

Param	Type	Description
angle	number Array.<number>	A number representing an angle in radians.

Example

```
sin(0) // 0
sin(1) // 0.8414709848078965
```

9.3.69. sort

sort(list) ⇒ Array.<any>

Description

This function accepts an array values and returns an array with the elements in sorted order. If there are mixed data types, the values will be grouped in order: numbers, strings, booleans, nulls. String sorting is based on code points and is not locale-sensitive. If the sort encounters any objects or arrays, it will throw an evaluation error.

Returns

Array.<any> - The ordered result

Parameters

Param	Type	Description
list	Array.<any>	to be sorted

Example

```
sort([1, 2, 4, 3, 1]) // returns [1, 1, 2, 3, 4]
sort(["20", 20, true(), "100", null(), 100]) // returns [20, 100, "100", "20", true, null]
```

9.3.70. sortBy

sortBy(elements, expr) ⇒ array

Description

Sort an array using an expression to find the sort key. For each element in the array, the expression is applied and the resulting value is used as the sort value. If the result of evaluating the expression against the current array element results in type other than a number or a string, a [TypeError](#) will occur.

Returns

array - The sorted array

Parameters

Param	Type	Description
elements	array	Array to be sorted
expr	expression	The comparison expression

Example

```
// returns ["e", "def", "abcd"]
sortBy(["abcd", "e", "def"], &length(@))

// returns [{year: 1910}, {year: 2010}, {year: 2020}]
sortBy([{year: 2010}, {year: 2020}, {year: 1910}], &year)

// returns [5, -10, -11, -15, 30]
sortBy([-15, 30, -10, -11, 5], &abs(@))
```

9.3.71. split

split(string, separator) ⇒ Array.<string> | Array.<Array.<string>>

Description

Split a string into an array, given a separator

Returns

Array.<string> | Array.<Array.<string>> - The array of separated strings

Parameters

Param	Type	Description
string	string Array.<string>	string to split
separator	string Array.<string>	separator where the split(s) should occur

Example

```
split("abcdef", "") // returns ["a", "b", "c", "d", "e", "f"]
split("abcdef", "e") // returns ["abcd", "f"]
```

9.3.72. sqrt

sqrt(num) ⇒ number | Array.<number>

Description

Find the square root of a number

Returns

number | Array.<number> - The calculated square root value

Parameters

Param	Type	Description
num	number Array.<number>	source number

Example

```
sqrt(4) // returns 2
```

9.3.73. startsWith

startsWith(subject, prefix) ⇒ boolean | Array.<boolean>

Description

Determine if a string starts with a prefix.

Returns

boolean | Array.<boolean> - true if **prefix** matches the start of **subject**

Parameters

Param	Type	Description
subject	string Array.<string>	string to search
prefix	string Array.<string>	prefix to search for

Example

```
startsWith("jack is at home", "jack") // returns true
```

9.3.74. stdev

stdev(values) ⇒ number

Description

Estimates standard deviation based on a sample. **stdev** assumes that its arguments are a sample of the entire population. If your data represents a entire population, then compute the standard deviation using **stdevp**. Non-numeric values (text, boolean, null etc) are ignored. If there are nested arrays, they are flattened.

Returns

number - [Standard deviation](#)

Parameters

Param	Type	Description
values	Array.<any>	The array containing numbers comprising the population. Array size must be greater than 1.

Example

```
stdev([1345, 1301, 1368]) // returns 34.044089061098404  
stdevp([1345, 1301, 1368]) // returns 27.797
```

9.3.75. stdevA

stdevA(numbers) ⇒ number

Description

Estimates standard deviation based on a sample. **stdev** assumes that its arguments are a sample of the entire population. If your data represents a entire population, then compute the standard deviation using **stdevpA**. Nested arrays are flattened. Null values are ignored. All other parameters are converted to number. If conversion to number fails, a type error is thrown.

Returns

number - [Standard deviation](#)

Parameters

Param	Type	Description
numbers	Array.<number>	The array of numbers comprising the population. Array size must be greater than 1.

Example

```
stdevA([1345, "1301", 1368]) // returns 34.044089061098404  
stdevpA([1345, 1301, "1368"]) // returns 27.797
```


9.3.76. stdevp

stdevp(values) ⇒ number

Description

Calculates standard deviation based on the entire population given as arguments. **stdevp** assumes that its arguments are the entire population. If your data represents a sample of the population, then compute the standard deviation using [stdev](#). Non-numeric values (text, boolean, null etc) are ignored. If there are nested arrays, they are flattened.

Returns

number - Calculated standard deviation

Parameters

Param	Type	Description
values	Array.<any>	The array containing numbers comprising the population. An empty array is not allowed.

Example

```
stdevp([1345, 1301, 1368]) // returns 27.797
stdev([1345, 1301, 1368]) // returns 34.044
```

9.3.77. stdevp

stdevp(numbers) ⇒ number

Description

Calculates standard deviation based on the entire population given as arguments. **stdevpA** assumes that its arguments are the entire population. If your data represents a sample of the population, then compute the standard deviation using [\[stdevA\]](#). Nested arrays are flattened. Null values are ignored. All other parameters are converted to number. If conversion to number fails, a type error is thrown.

Returns

number - Calculated standard deviation

Parameters

Param	Type	Description
numbers	Array.<number>	The array of numbers comprising the population. An empty array is not allowed.

Example

```
stdevpA([1345, "1301", 1368]) // returns 27.797
stdevA([1345, 1301, "1368"]) // returns 34.044
```

9.3.78. substitute

substitute(text, old, new, [which]) ⇒ string | Array.<string>

Description

Generates a string from the input **text**, with text **old** replaced by text **new** (when searching from the left). If there is no match, or if **old** has length 0, **text** is returned unchanged. Note that **old** and **new** may have different lengths.

Returns

string | Array.<string> - replaced string

Parameters

Param	Type	Description
text	string Array.<string>	The text for which to substitute code points.
old	string Array.<string>	The text to replace.
new	string Array.<string>	The text to replace old with. If new is an empty string, then occurrences of old are removed from text .
[which]	integer Array.<integer>	The zero-based occurrence of old text to replace with new text. If which parameter is omitted, every occurrence of old is replaced with new .

Example

```
substitute("Sales Data", "Sales", "Cost") // returns "Cost Data"
substitute("Quarter 1, 2001", "1", "2", 1) // returns "Quarter 1, 2002"
substitute("Quarter 1, 2011", "1", "2", 2) // returns "Quarter 1, 2012"
```

9.3.79. sum

sum(collection) ⇒ number

Description

Calculates the sum of the provided array. An empty array will produce a return value of 0. Any values that are not numbers (null, boolean, strings, objects) will be ignored. If any parameters are arrays, the arrays will be flattened.

Returns

number - The computed sum

Parameters

Param	Type	Description
collection	Array.<any>	array of values

Example

```
sum([1, 2, 3]) // returns 6
```

9.3.80. tan

tan(angle) ⇒ number | Array.<number>

Description

Computes the tangent of a number in radians

Returns

number | Array.<number> - The tangent of **angle**

Parameters

Param	Type	Description
angle	number Array.<number>	A number representing an angle in radians.

Example

```
tan(0) // 0
tan(1) // 1.5574077246549023
```

9.3.81. time

time(hours, [minutes], [seconds]) ⇒ number

Description

Construct and returns a [time value](#). If any of the units are greater or less than their normal range, the overflow/underflow will be added/subtracted from the next greater unit.

Returns

number - Returns a date/time value representing the fraction of the day consumed by the given time

Parameters

Param	Type	Default	Description
hours	integer		Zero-based integer value between 0 and 23 representing the hour of the day.
[minutes]	integer	0	Zero-based integer value representing the minute segment of a time.
[seconds]	integer	0	Zero-based integer value representing the seconds segment of a time.

Example

```
time(12, 0, 0) | [hour(@), minute(@), second(@)] // returns [12, 0, 0]
```

9.3.82. toArray

toArray(arg) ⇒ array

Description

Converts the provided argument to an array. The conversion happens as per the following rules:

- array - Returns the provided value.
- number/string/object/boolean/null - Returns a one element array containing the argument.

Returns

array - The resulting array

Parameters

Param	Type	Description
arg	any	parameter to turn into an array

Example

```
toArray(1) // returns [1]
toArray(null()) // returns ['null']
```

9.3.83. toDate

toDate(ISOString) ⇒ number

Description

Converts the provided string to a date/time value.

Returns

number - The resulting [date/time number](#). If conversion fails, return null.

Parameters

Param	Type	Description
ISOString	string	An ISO8601 formatted string . (limited to the RFC 3339 profile) If the string does not include a timezone offset (or trailing 'Z'), it will be assumed to be local time

Example

```
toDate("20231110T130000+04:00") // returns 19671.375
toDate("2023-11-10T13:00:00+04:00") // returns 19671.375
toDate("20231110") | year(@) & "/" & month(@) // returns "2023/11"
```

9.3.84. today

today() ⇒ number

Description

Returns a [date/time](#) value representing the start of the current day. i.e. midnight

Returns

number - today at midnight

9.3.85. toNumber

toNumber(value, [base]) ⇒ number

Description

Converts the provided arg to a number. The conversions follow the [type coercion rules](#) but will also:

- * Convert non-numeric strings to zero
- * Convert arrays to arrays of numbers

Returns

number - The resulting number. If conversion to number fails, return null.

Parameters

Param	Type	Default	Description
value	any Array.<any>		to convert to number
[base]	integer Array.<integer>	10	If the input arg is a string, the base to use to convert to number. One of: 2, 8, 10, 16. Defaults to 10.

Example

```
toNumber(1) // returns 1
toNumber("10") // returns 10
toNumber({a: 1}) // returns null
```

```
toNumber(true()) // returns 1
toNumber("10f") // returns null
toNumber("FF", 16) // returns 255
```

9.3.86. toString

toString(arg, [indent]) ⇒ string

Description

Returns the argument converted to a string. If the argument is a string, it will be returned unchanged. Otherwise, returns the JSON encoded value of the argument.

Returns

string - The result string.

Parameters

Param	Type	Default	Description
arg	any		Value to be converted to a string
[indent]	integer	0	Indentation to use when converting objects and arrays to a JSON string

Example

```
toString(1) // returns "1"
toString(true()) // returns "true"
toString({sum: 12 + 13}) // '{"sum":25}'
toString("hello") // returns "hello"
```

9.3.87. trim

trim(text) ⇒ string | Array.<string>

Description

Remove leading and trailing spaces (U+0020), and replace all internal multiple spaces with a single space. Note that other whitespace characters are left intact.

Returns

string | Array.<string> - trimmed string

Parameters

Param	Type	Description
text	string Array.<string>	string to trim

Example

```
trim("  ab  c  ") // returns "ab c"
```

9.3.88. true

true() ⇒ boolean

Description

Return constant boolean true value. Expressions may also use the JSON literal: `\ 'true\ '`

Returns

boolean - True

9.3.89. trunc

trunc(numA, [numB]) ⇒ number | Array.<number>

Description

Truncates a number to an integer by removing the fractional part of the number. i.e. it rounds towards zero.

Returns

number | Array.<number> - Truncated value

Parameters

Param	Type	Default	Description
numA	number Array.<number>		number to truncate
[numB]	integer Array.<integer>	0	A number specifying the number of decimal digits to preserve.

Example

```
trunc(8.9) // returns 8
trunc(-8.9) // returns -8
trunc(8.912, 2) // returns 8.91
```

9.3.90. type

type(subject) ⇒ string

Description

Finds the type name of the given **subject** argument as a string value. The return value will be one of the following:

- number
- string
- boolean
- array
- object
- null

Returns

string - The type name

Parameters

Param	Type	Description
subject	any	type to evaluate

Example

```
type(1) // returns "number"
type("") // returns "string"
```

9.3.91. unique

unique(input) ⇒ array

Description

Find the set of unique elements within an array

Returns

array - array with duplicate elements removed

Parameters

Param	Type	Description
input	array	input array

Example

```
unique([1, 2, 3, 4, 1, 1, 2]) // returns [1, 2, 3, 4]
```

9.3.92. upper

upper(input) ⇒ string | Array.<string>

Description

Converts all the alphabetic code points in a string to uppercase.

Returns

string | Array.<string> - the upper case value of the input string

Parameters

Param	Type	Description
input	string Array.<string>	input string

Example

```
upper("abcd") // returns "ABCD"
```

9.3.93. value

value(subject, index) ⇒ any

Description

Perform an indexed lookup on an object or array

Returns

any - the result of the lookup – or **null** if not found.

Parameters

Param	Type	Description
subject	object array null	on which to perform the lookup. When querying for hidden properties , subject may be any data type.
index	string integer	if subject is an object, index must be a string indicating the key name to search for. If subject is an array, then index must be an integer indicating the offset into the array

Example

```
value({a: 1, b:2, c:3}, "a") // returns 1  
value([1, 2, 3, 4], 2) // returns 3
```

9.3.94. values

values(obj) ⇒ array

Description

Generates an array of the values of the provided object. Note that because JSON objects are inherently unordered, the values associated with the provided object are also unordered.

Returns

array - array of the values

Parameters

Param	Type	Description
obj	object	source object

Example

```
values({a : 3, b : 4}) // returns [3, 4]
```

9.3.95. weekday

weekday(date, [returnType]) ⇒ integer | Array.<integer>

Description

Extract the day of the week from a date. The specific numbering of the day of week is controlled by the **returnType** parameter:

- 1 : Sunday (1), Monday (2), ..., Saturday (7)
- 2 : Monday (1), Tuesday (2), ..., Sunday(7)
- 3 : Monday (0), Tuesday (1), ..., Sunday(6)

Returns

integer | Array.<integer> - day of the week

Parameters

Param	Type	Default	Description
date	number Array.<number>		date/time value for which the day of the week is to be returned. Date/time values can be generated using the datetime , toDate , today , now and time functions.
[returnType]	integer Array.<integer>	1	Determines the representation of the result. An unrecognized returnType will result in a error.

Example

```
weekday(datetime(2006,5,21)) // 1
weekday(datetime(2006,5,21), 2) // 7
weekday(datetime(2006,5,21), 3) // 6
```

9.3.96. year

year(date) ⇒ integer | Array.<integer>

Description

Finds the year of a datetime value

Returns

integer | Array.<integer> - The year value

Parameters

Param	Type	Description
date	number Array.<number>	input date/time value Date/time values can be generated using the datetime , toDate , today , now and time functions.

Example

```
year(datetime(2008,5,23)) // returns 2008
```

9.3.97. zip

zip(...arrays) ⇒ array

Description

Generates a convolved (zipped) array containing grouped arrays of values from the array arguments from index 0, 1, 2, etc. This function accepts a variable number of arguments. The length of the returned array is equal to the length of the shortest array.

Returns

array - An array of arrays with elements zipped together

Parameters

Param	Type	Description
...arrays	array	array of arrays to zip together

Example

```
zip([1, 2, 3], [4, 5, 6, 7]) // returns [[1, 4], [2, 5], [3, 6]]
```

10. Integrations

The json-formula API allows integrations to customize various json-formula behaviors.

10.1. Globals

By default, json-formula has one global symbol: `@`. A host may inject additional global identifiers. These identifiers must be prefixed with the dollar (`$`) symbol.

Examples

Given: a global symbol:

```
{
  "$days": [
    "Monday", "Tuesday", "Wednesday", "Thursday", "Friday", "Saturday",
    "Sunday"
  ]
}
```

```
eval(value($days, weekday(datetime(date.year, date.month, date.day), 3)),
{
  "date": {
    "year": 2023,
    "month": 9,
    "day": 13
  }
}) -> "Wednesday"
```

10.2. Specify locale

The default locale for json-formula is `en-US`. A host may specify an alternate locale. The locale setting affects only the behavior of the `casefold()` function.

10.3. Custom toNumber

In various contexts, json-formula converts values to numbers. A host may provide its own `toNumber()` function that json-formula will use in place of the default functionality. For example, a custom `toNumber()` could make use of locale-specific date formats to attempt to convert a string to a date value, or could allow currency values e.g., "\$123.45" to be converted to number.

10.4. Additional Functions

A host may provide its own set of functions to augment the base set provided by json-formula.

10.5. Hidden Properties

A host system may construct its source JSON data with complex properties that have nested structure that can be found through explicit navigation, but will not be found through normal tree searching. Here is an example of how this can be configured in JavaScript:

```
function createField(id, value) {
  class Field {
    valueOf() { return value; }

    toString() { return value.toString(); }

    toJSON() { return value; }
  }
  const f = new Field();
  Object.defineProperty(f, '$id', { get: () -> id });
  Object.defineProperty(f, '$value', { get: () -> value });

  return f;
}

const json = {
  "street": createField("abc123", "Maple Street"),
  "city": createField("def456", "New York")
}
```

Given this configuration, these search results are possible:

```
street -> "Maple Street"
street.$value -> "Maple Street"
street.$id -> "abc123"
type(street) -> "string"
keys(street) -> []
```

10.6. Tracking

A host system may want to track which properties are accessed during the evaluation of an expression. This can be done by providing a **track** function on the object being evaluated. The **track** function will be called with the object being evaluated and the key being accessed. Here is an example of how this can be configured in JavaScript:

```
properties[Symbol.for('track')] = (obj, key) => trackDependent(obj, key);
```